



ITU ML REPORT

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OBJECTIVE FUNCTION

Find a Machine Learning Process to accurately predict downlink or uplink

Criteria to be evaluated on:

1. Total data set features used
2. R^2 value
3. Generated report

Note that the dataset would be split by a target and source domain.

Eg. By operator, or by cars

APPROACH

FEATURE SELECTION

Variables that affect downlink or uplink

DOMAIN SELECTION

Splitting of the data set

ALGORITHM ANALYSIS

Determining how effective the algorithm is

REPORT GENERATION

Finalizing the algorithm and explaining feature and domain choices

MAIN DATA COMPONENTS

PCell

Variables led by “PCELL”

Primary data results for variables such as:
RSRP, RSRQ, and SINR

SCell

Variables led by “SCELL”

Secondary data results for variables such as:
RSRP, RSRQ, and SINR

Misc

Variables not included in PCELL or SCELL

Data results for variables such as:
ping, altitude, speed

FEATURE SELECTION

SELECT FROM THE KNOWN

Choose features that are known to affect downlink data rates

Examine the affect – positive or negative – on the machine learning algorithm

Once a set of known variables has proven to increase efficacy of algorithm, start on:

TRIAL AND ERROR ON THE UNKNOWN

Choose variables that may not necessarily affect downlink data rate and using trial and error, cull the list to those that can help our algorithm

EXAMPLE FEATURE SELECTION

No.	Features selected	Result	No.	Features selected	Result
1	Pcell: RSRP, RSRQ, RSSI, SNR	MAPE: 102.81 % Mean Squared Error: 748.9089965820312 R-Squared Value: 0.509003524825407 Mean Absolute Error (MAE): 19.57 Mean Absolute Percentage Error (MAPE): 102.81%	2	Pcell: RSRP, RSRQ, RSSI, SNR, Uplink Tx Power, Downlink Transport Block Size	MAPE: 232.68% Mean Squared Error: 829.4603271484375 R-Squared Value: 0.45293608529149754 Mean Absolute Error (MAE): 22.22 Mean Absolute Percentage Error (MAPE): 232.68%
No.	Features selected	Result	No.	Features selected	Result
3	PCell: RSRP, RSRQ, RSSI, SNR, Downlink Average MCS SCell: RSRP, RSRQ, RSSI, SNR, Downlink Average MCS Misc: ping_ms	MAPE: 9.15% Mean Squared Error: 128.09872436523438 R-Squared Value: 0.94543727931438 Mean Absolute Error (MAE): 6.15 Mean Absolute Percentage Error (MAPE): 9.15%	4	PCell: RSRP, RSRQ, RSSI, SNR, Downlink Average MCS, frequency (MHz) SCell: RSRP, RSRQ, RSSI, SNR, Downlink Average MCS, frequency (MHz) Misc: ping_ms	MAPE: 9.18% Mean Squared Error: 89.81575775146484 R-Squared Value: 0.9617436276875649 Mean Absolute Error (MAE): 5.85 Mean Absolute Percentage Error (MAPE): 9.18%



FEATURE SELECTION

Unrelated Features

Now that a high r^2 value has been achieved, we will add “fluff” features to increase overall score

Different Models

Changing hidden network size and run time in artificial neural networks, or trying out different regression models

Different Train and Test Set Sizes

Different percentages of train/set set to see the effect it had on the algorithm

Strategy Specific Selection

Adding specific “fluff” features

Choosing features that had little to no negative effects on algorithm

DOMAINS – TRAIN AND TEST SETS

Park and Avenue

Data collected around parks is used to predict downlink around avenues

Operator 1 and Operator 2

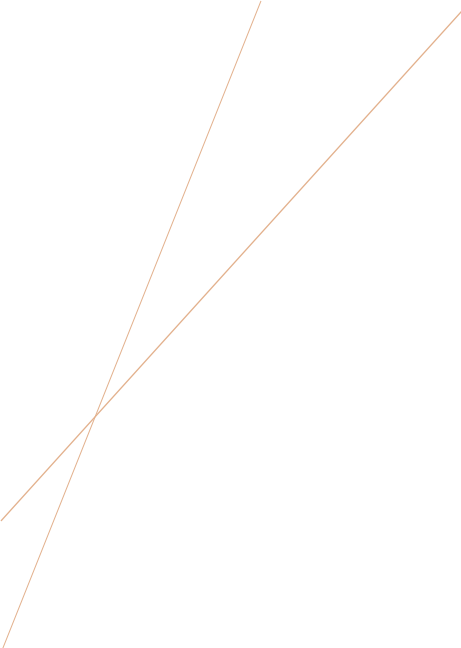
Data collected by operator 1 is used to predict the downlink of operator 2

Uplink and Downlink

Data collected during uplink is used to predict downlink

Highway and Residential

Data collected on highways is used to predict downlink in residential areas



FINALIZING ALGORITHM

PARK AND AVENUE

It was determined through testing that the park and avenue domains were best for our algorithm

Possibly due to similar levels of interference or other factors

EVALUATING R^2

Sharp decrease in accuracy when switching train and test sets

Could be explained by the size of train and test sets changing

FUTURE PLANS

To improve the algorithm some steps may be taken such as:

Adding more features, training the algorithm longer, changing domain

A series of thin, light-brown lines forming an abstract geometric pattern in the top-left corner of the slide. The lines intersect to create various triangular and polygonal shapes.

THANK YOU

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https://github.com/dkapur2026/ITU_AI_ML_CHALLENGE/tree/main

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